

McMaster University

Investigating Intracortical Myelin as a Biomarker for Depression Severity in Mood Disorders



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HYPOTHESES

H_0 : There is no correlation between R_1 (a surrogate marker of intracortical myelin [ICM]) and MADRS score (depression severity).

H_a : There is a correlation between R_1 and MADRS score.

METHODS

Clinical Data Collection

- Participants ($n = 103$) — healthy controls ($n = 10$; HC), Depressive Disorder ($n = 83$; MDD or PDD), and Bipolar Disorder ($n = 10$; BD) — were recruited through the ENABLE master clinical trial platform. Clinical data collection focused on assessing depression severity through collection of the clinical rated 10-item **Montgomery-Åsberg Depression Rating Scale (MADRS)**.

MRI Acquisition

- High-resolution quantitative MRI (qMRI) data were collected on a 3 Tesla GE scanner at St. Joseph's Healthcare in Hamilton, Ontario and at the University of Calgary in Calgary, Alberta
- A 3D inversion recovery gradient-echo sequence and a 3D gradient-echo sequence without inversion were used to generate longitudinal relaxation rate (R_1) maps.

Cortical Surface Reconstruction

- Anatomical images were processed using **FreeSurfer (v8.1.0)** to generate pial and white matter surfaces in MNI-152 space. These surfaces were then used within **Connectome Workbench (v2.1.0)** to reconstruct a middle-depth cortical surface for R_1 mapping.

Cortical Parcellation

- The **Human Connectome Project Multi-Modal Parcellation (HCP MMP) atlas¹** was used because this atlas takes cortical myeloarchitecture into account in order to define 360 cortical regions of interest (ROIs).

Statistical Approach

- Statistical significance was assessed based on a **multiple linear regression model** using the General Linear Model (GLM) below:

$$R_1 \sim \text{age} + \text{age}^2 + \text{sex} + \text{MADRS}$$

RESULTS

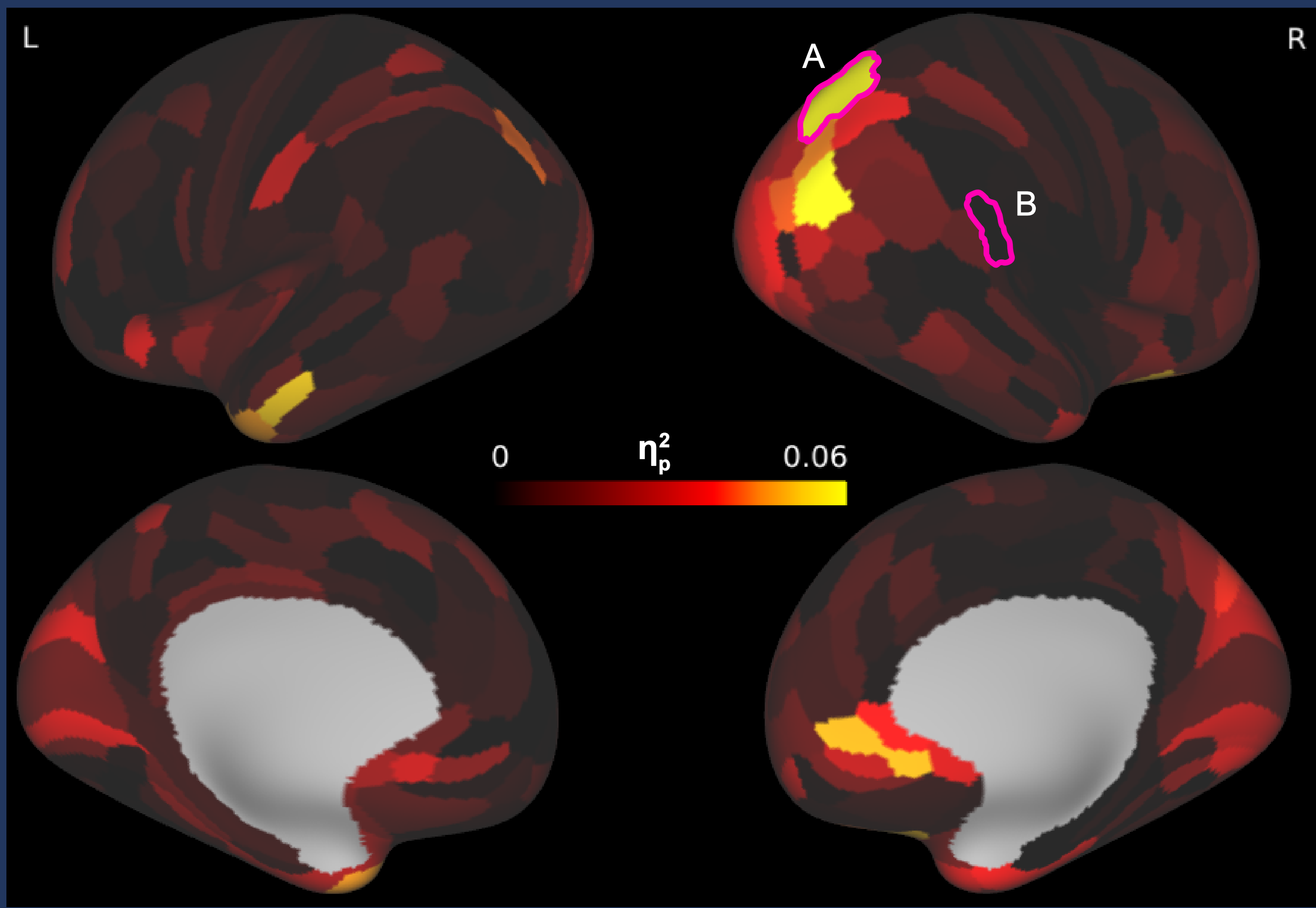


Figure 1. Inflated cortical surface reconstruction with η_p^2 values plotted across all 360 regions of interest (ROIs). Areas colored in orange or yellow demonstrate medium to large effect sizes ($\eta_p^2 > 0.06$), as measured by η_p^2 . (A) $\eta_p^2 = 0.099$ Right Intraparietal Sulcus Area (R_IPS1) (B) $\eta_p^2 = 0$ Right Perisylvian Language Area (R_PSL)

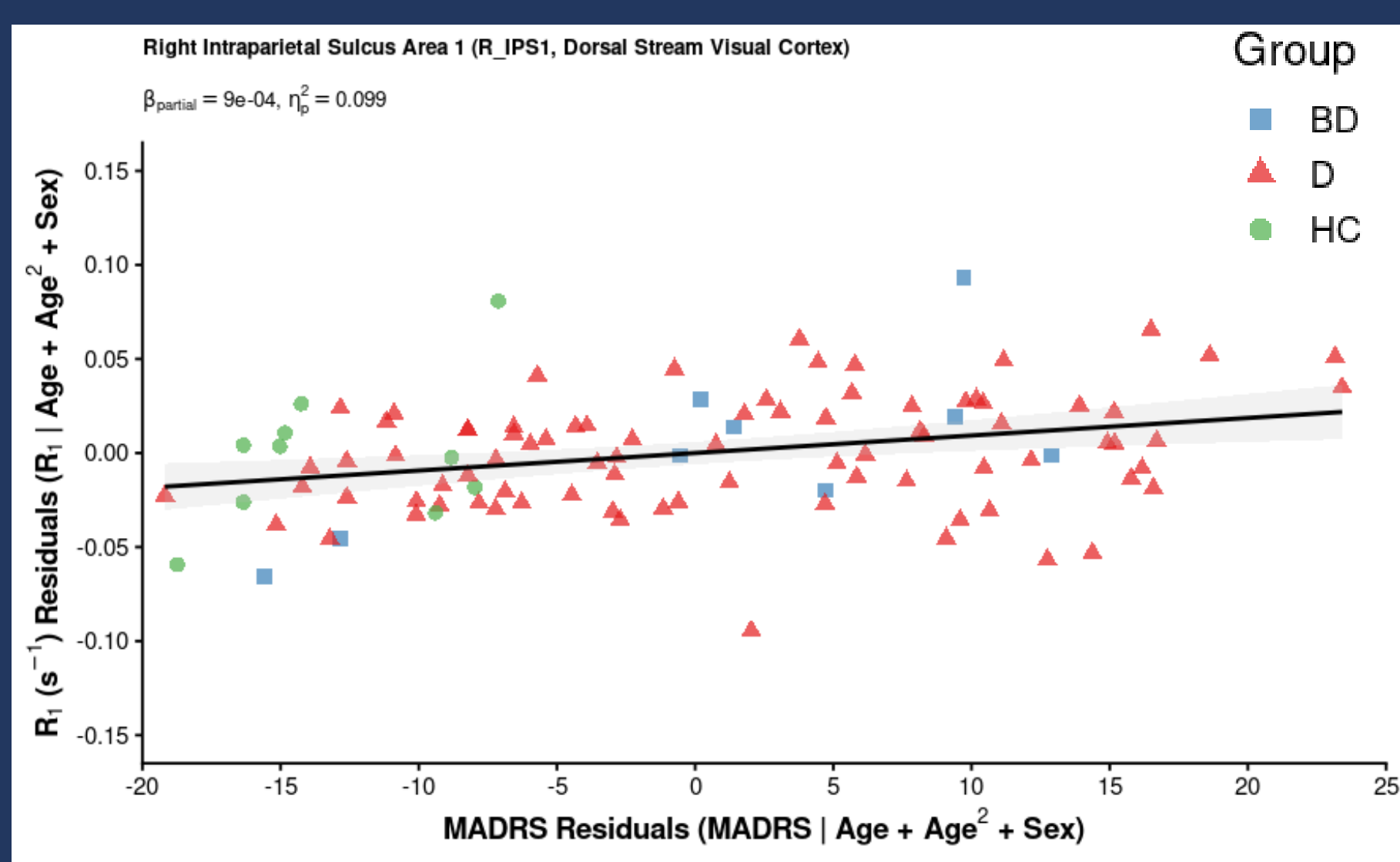


Figure 2. Relationship between R_1 and MADRS. Partial regression plot showing the **positive correlation** between R_1 and MADRS score in the Right Intraparietal Sulcus Area 1 (R_IPS1).

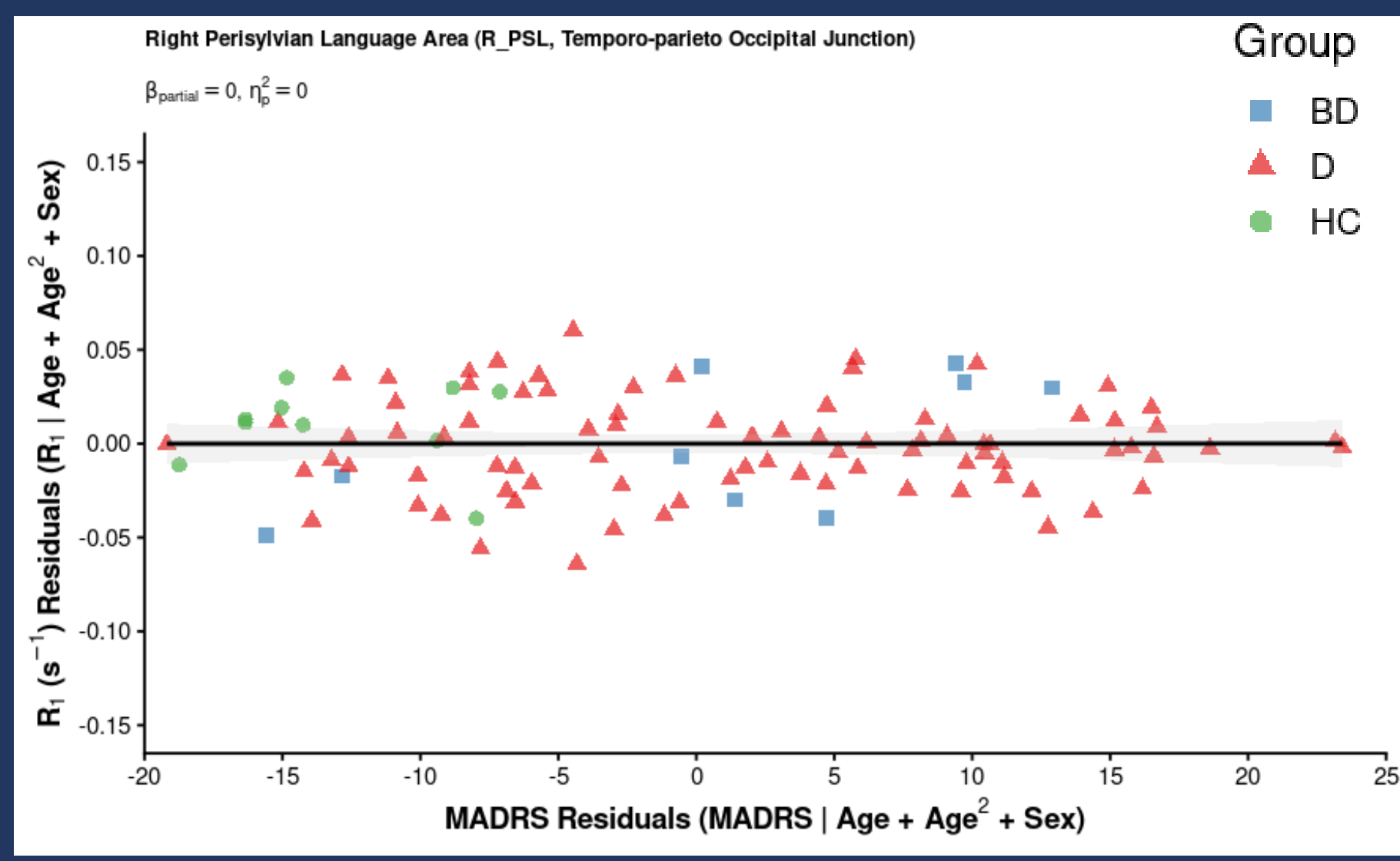


Figure 3. Relationship between R_1 and MADRS. Partial regression plot showing the **lack of correlation** between R_1 and MADRS score in the Right Perisylvian Language Area (R_PSL).

CONCLUSIONS

- For the Right IPS1 region **MADRS score can explain ~10% of the variance in R_1** after taking age and sex into consideration. However, MADRS showed no relationship with R_1 across many ROIs, such as the Right PSL.
- The results of this analysis show that there are **a cluster of ROIs related to visual processing in the right hemisphere (Figure 1)** that have moderate effect size values.
- This finding implicates visual cortical areas in depression, a finding that has been investigated in both mouse and human research on depression^{2,3}. These studies suggest that visual areas and pathways may serve as potential targets for therapeutic intervention², be associated with depressive behaviours³, and serve as predictors for therapeutic responsiveness⁴.
- However, further research is required to understand the precise nature of the correlation between ICM, in cortical ROIs related to visual processing, and depression severity.

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